

SEQUENCE & SERIES



Ahh! AA

MCQ # 01 The last term of a sequence given by $T_n = n^2 - 14n$ is 51. The no. of terms in the sequence is?

- (a) 15 (b) 16 (c) 17 (d) 18

$$T_n = n^2 - 14n$$

$$51 = n^2 - 14n$$

$$0 = n^2 - 14n - 51$$

$$0 = n^2 - 17n + 3n - 51$$

$$= n(n-17) + 3(n-17) = (n-17)(n+3)$$

$$n-17=0$$

$$\boxed{n=17}$$

$$n+3=0$$

$$\boxed{n=-3} \times$$

↳ n can't be negative.

no of terms can't be negative.

For n Arithmetic Means =

$$A_n = a + nd \text{ where } d = \frac{b-a}{n+1}$$

Q# 01 Find two Arithmetic mean between a & b.

(a) $\frac{3a+b}{3}, \frac{3b+a}{3}$

(b) $\frac{2a+b}{3}, \frac{2b+a}{3}$

(c) $\frac{2b+a}{3}, \frac{2a+b}{3}$

(d) $\frac{3b+a}{3}, \frac{3a+b}{3}$

$$A_1 = a + d$$

$$A_2 = a + 2d$$

$$d = \frac{b-a}{n+1} \Rightarrow \frac{b-a}{2+1} = \frac{b-a}{3}$$

$$A_1 = a + \left(\frac{b-a}{3}\right) = \frac{3a+b-a}{3} = \frac{2a+b}{3}$$

$$A_2 = a + \frac{2b-2a}{3} = \frac{3a+2b-2a}{3} = \frac{a+2b}{3} = \frac{2b+a}{3}$$

↳ options are reversed

here like $(x, y) = (2, 3)$

you can't write it as $(x, y) = (3, 2)$

For n no of Geometric Mean:

$$G_n = ar^n \text{ where } r = \left[\frac{b}{a} \right]^{\frac{1}{n+1}}$$

Arithmetic Progression:

Q#(1) If the p^{th} term of an A.P is "P", the q^{th} term is "q" then $(p+q)^{\text{th}}$ term will be?

- (a) P (b) q (c) P+q (d) P-q

$$T_p \Rightarrow P$$

$$T_q \Rightarrow q$$

$$T_{p+q} \Rightarrow P+q$$

Q#(2) The T_n of the Sequence $\frac{1}{3}, 1\frac{2}{3}, 3, \dots$ is.

- (a) $\frac{1}{3}(4n-3)$ (b) $\frac{1}{3}(4n-2)$ (c) $\frac{1}{3}(4n-5)$ (d) $(4n-3)$

Q#(3) The angle of θ for which $\sin \theta, 2\cos \theta, 2\sin \theta$ respectively are the first three terms of an A.P is.

- (a) 53.1° (b) 30.1° (c) 45.1° (d) 90.1°

$$2\sin \theta - 2\cos \theta = 2\cos \theta - \sin \theta$$

$$2\sin \theta + \sin \theta = 2\cos \theta + 2\cos \theta$$

$$3\sin \theta = 4\cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = \frac{4}{3}$$

$$\tan \theta = \frac{4}{3}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = \tan^{-1}(1.3)$$

$$\theta = \tan^{-1}(1) = 45^\circ$$

Q#04) The 10th term of an A.P is "q" & qth term is "10". Then what is the value of T_{10+q} ?

$$T_{10} = q, \quad T_q = 10$$

$$T_{10+q} = ?$$

Note if in an A.P the values of two terms are label of each other then ~~their sum term value~~ will be equal to zero.

- (a) p (b) q (c) 0 (d) q+10

Q#05) The 10th term of an H.P is "21" & 21st term is "10". Find the 210th term.

- (a) 0 (b) 1 (c) -1 (d) 10

$$T_{10} = 21$$

$$T_{21} = 10$$

$$T_{210} = ?$$

Note if in an H.P two terms are label of each other their product term value will be equal to 1

Q#06) The 10th term of an A.P is 5 & 5th term is 10, then value of T_{50} ?

- (a) 0 (b) 1 (c) -35 (d) -1

Note if terms are label of each other in A.P & their product term is asked then the value of product term will be ~~difference of~~ equal to the difference of addition of values & multiplication of values (a) - (x)

$$T_{50} = (5+10) - (5 \times 10) = 15 - 50 = -35.$$

Q#07) The sum of first n terms of two A.P are the ratio $3n+31 : Sn-3$. The ratio of their 9th term is-

- (a) 1:1 (b) 1:2 (c) 2:1 (d) 1:3

$$S_n = 3n+31$$

$$S_n' = Sn-3$$

$$T_9 = ?$$

$$T_9'$$

Shortcut for finding no of terms $n = (\text{Term no} \times 2) - 1$

$$n = (9 \times 2) - 1 = 17$$

$$S_n = \frac{3(17) + 31}{5(17) - 3} = 1:1$$

$$\frac{T_9}{T_9'} = \frac{1}{1}$$

Q# 68) if $1+3+5+\dots$ n terms $= 0.95$ then the value of n = ?
 $2+4+6+\dots$ n terms

$$\text{Even no} \Rightarrow n(n+1)$$

$$\text{odd no} \Rightarrow n^2$$

$$\frac{n^2}{n(n+1)} = \frac{95}{100}$$

$$100n = 95n + 95$$

$$5n = 95$$

$$\boxed{n = 19}$$

Q# 69) How many terms of the series $-8, -6, -4, \dots$ make the sum 90?
 $S_n = \frac{n}{2} \{ 2a + (n-1)d \}$

$$90 = \frac{n}{2} \{ 2(-8) + (n-1)(+2) \}$$

$$90 = \frac{n}{2} (2) \{ -8 + n - 1 \}$$

$$90 = n(-9 + n) \rightarrow \text{you can back solve}$$

$$90 = -9n + n^2 \quad \text{from here also}$$

$$0 = n^2 - 9n - 90$$

$$0 = n^2 - 15n + 6n - 90$$

$$0 = n(n-15) + 6(n-15)$$

$$(n-15)(n+6) = 0$$

$$\boxed{n = 15}$$

2	90
5	45
3	9
3	3
	1

Q# 10) The n^{th} term of an A.P. is given by $T_n = 7n - 5$. The sum of first 16 terms will be.

(A) 862 (B) 872 (C) 882 (D) 892

$$S_n = \frac{n}{2} \{ a + l \} \quad | \quad T_1 = 7 - 5 = 2 \quad | \quad T_{16} = 112 - 5 = 107$$

$$S_n = 8 \{ 2 + 107 \} = 109(8) = 872$$

Q#11 The Sum to n -terms of an A.P. is given by the formula $S_n = 2n^2 + n$. The first term & common difference.

- (a) 1, 2 (b) 2, 2 (c) 2, 3 (d) 3, 4

$$S_n = n/2 \{ 2a_1 + (n-1)d \}$$

$$S_n = 2n^2 + n$$

$$2n^2 + n = n/2 \{ 2a_1 + (n-1)d \}$$

$$4n^2 + 2n = n \{ 2a_1 + (n-1)d \}$$

$$2n(n+1) = n \{ 2a_1 + (n-1)d \}$$

$$4n+2 = 2a_1 + (n-1)d$$

For a_1 put $n=1$

$$4+2 = 2a_1 + 0$$

$$6 = 2a_1$$

$$\boxed{a_1 = 3}$$

For d put $n=2$ you can't put any value of n

$$4(2)+2 = 2(3) + (2-1)d$$

$$8+2 = 6+d$$

$$10 = 6+d$$

$$d = 10 - 6 = 4$$

$$\boxed{d = 4}$$

Q#12 First two terms of an H.P. are $2/39$ & $2/37$. The 40th term will be?

- (a) $2/39$ (b) $-2/39$ (c) $2/37$ (d) $-2/37$

$$A.P. \Rightarrow T_1 = 39/2, T_2 = 37/2$$

$$a_2 = a_1 + (n-1)d$$

$$37/2 = 39/2 + d$$

$$\boxed{d = -1}$$

$$a_{40} = \frac{39 + 39(-1)}{2}$$

$$a_{40} = \frac{39 - 39}{2} = \frac{39 - 78}{2} = -\frac{39}{2}$$

$$\therefore \text{In H.P.} = -2/39$$

Q#13 Which term of the G.P $8^{-1}, 4^{-1}, 2^{-1}, 1, \dots$ is 32?

(A) 6

(B) 8

(C) 7

(D) 9

$$a_n = ar^{n-1}$$

$$32 a_{n-1} = \frac{1}{8} (2)^{n-1}$$

$$256 = 2^{n-1}$$

$$256 = 2^{n-1}$$

$$2^8 = 2^{n-1}$$

$$n-1 = 8$$

$$\boxed{n = 9}$$

Q#14 The 20th term of the Series $64 - 16 + 4 - 1 + \dots$ is?

(A) $\frac{1}{4^{16}}$

(B) $-\frac{1}{4^{16}}$

(C) $\frac{1}{2^{16}}$

(D) $-\frac{1}{2^{16}}$

(E) $\frac{1}{3^{14}}$

$$r = -\frac{1}{4}$$

$$a_{20} = 64 \left(-\frac{1}{4}\right)^{19}$$

$$a_{20} = 64 \left(-\frac{1}{4^{19}}\right)$$

$$a_{20} = \frac{4^3}{4^{19}}$$

$$a_{20} = -4^{3-19} = -4^{16} = -\frac{1}{4^{16}}$$

Q#15 The least no. of terms of the G.P $4, 8, 16, \dots$ that must be taken for their sum to exceed 500.

(A) 3

(B) 4

(C) 5

(D) 6

(E) 7

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \left| \quad 500 = \frac{4(2^n - 1)}{2 - 1} \quad \left| \quad 500 = \frac{4(2^n - 1)}{4} \right. \right.$$

$$125 = 2^n - 1$$

$$126 = 2^n$$

$$2^7 = 2^n$$

$$\boxed{n = 7}$$

Q#16 The ratio of the Sum of the first four terms of the G.P to the Sum of the first eight terms is $81/97$. Find the common ratio.

- (a) $2/3$ (b) $-2/3$ (c) $3/2$ (d) $-3/2$ (e) both (a) & (b)

$$\frac{S_4}{S_8} = \frac{81}{97}$$

$$\frac{\frac{a_1(1-r^4)}{1-r}}{\frac{a_1(1-r^8)}{1-r}} = \frac{81}{97}$$

$$\frac{1-r^4}{1-r^8} = \frac{81}{97}$$

$$\frac{1-r^4}{1-(r^4)^2} = \frac{81}{97}$$

$$\frac{(1-r^4)}{(1-r^4)(1+r^4)} = \frac{81}{97}$$

$$\frac{1}{1+r^4} = \frac{81}{97}$$

$$97 = 81 + 81r^4$$

$$\frac{16}{81} = r^4$$

$$r = \pm 2/3$$

→ The $\pm$ sign automatically appears because both values satisfy the original equation

r^4 is same for $+r$ & $-r$

Q#17 The Sum of the first three terms of a G.P is 117 & the Sum of the next 3 terms is $13/3$, ratio will be?

- (a) $1/3$ (b) 3 (c) $2/3$ (d) $3/2$

$$a + ar + ar^2 = 117 \quad \text{--- (1)}$$

$$ar^3 + ar^4 + ar^5 = 13/3 \quad \text{--- (2)}$$

Divide (1) & (2)

$$\frac{ar^3 + ar^4 + ar^5}{a + ar + ar^2} = \frac{13/3}{117}$$

$$\frac{ar^3(1+r+r^2)}{a(1+r+r^2)} = \frac{13/3}{117}$$

$$r^3 = \frac{13/3}{3(117)}$$

$$r^3 = \frac{1}{27} \quad \left| \sqrt[3]{\frac{1}{27}} = \frac{1}{3} \right|$$

Q #18 The 2nd & 5th term of an G.P are 6 & 48 respectively. The Sum of first 10 terms will be?

$$\begin{array}{l} a_2 = 6 \\ a_5 = 48 \end{array}$$

$$\frac{ar^4}{ar} = \frac{48}{6}$$

$$r^3 = 8$$

$$\boxed{r = 2}$$

$$a(2) = 6$$

$$\boxed{a = 3}$$

$$S_{10} = \frac{3(2^{10}-1)}{2-1}$$

$$3(1023) = 3069$$

$$\begin{array}{r} 1023 \\ \times 3 \\ \hline 3069 \end{array}$$

Q #19 The third term of a convergent G.P is the arithmetic mean of the first & second term. If the first term is 1 the Sum to infinity will be?

(a) $\frac{1}{4}$

(b) $\frac{1}{3}$

(c) $\frac{1}{2}$

(d) $\frac{3}{2}$

(e) $\boxed{\frac{2}{3}}$

$$\begin{array}{l} a, ar, ar^2 \\ a=1, \quad ar^2 = \frac{a+ar}{2} \end{array}$$

$$ar^2 = \frac{a(1+r)}{2}$$

$$2r^2 = 1+r$$

$$2r^2 - r - 1 = 0$$

$$2r^2 - 2r + r - 1 = 0$$

$$2r(r-1) + 1(r-1) = 0$$

$$(r-1)(2r+1) = 0$$

$$r = 1, \quad r = -\frac{1}{2}$$

$$* S_{\infty} = \frac{1}{1 - (-\frac{1}{2})} = \frac{1}{1 + \frac{1}{2}}$$

$$S_{\infty} = \frac{1}{\frac{3}{2}}$$

$$\boxed{S_{\infty} = \frac{2}{3}}$$

As it is mentioned that Series is convergent so, r must be less than 1 so we will ignore 1 & take $-\frac{1}{2}$ *

Keep going!

$$\sum_{n=1}^{\infty} \frac{1}{5^n}$$

condition for convergence of a G.P Series

An infinite G.P Series converges if $|r| < 1$

$$\text{Here, } |r| = \frac{1}{5} < 1$$

condition satisfied!

& for a convergent Series Sum approaches a finite number

$$S_{\infty} = \frac{1}{4}$$

$$\textcircled{1} \# \textcircled{20} \quad 11^3 + 12^3 + 13^3 + \dots + 20^3 = ?$$

As particular formulas were only valid when first term is 1 but in this case first term is not one so,

$$\text{Sum}(1-20) - \text{Sum}(1-10) = \text{Sum}(11-20)$$

$$\sum_{n=1}^{20} n^3 - \sum_{n=1}^{10} n^3 = \sum_{n=11}^{20} n^3$$

$$\left[\frac{20[20+1]}{2} \right]^2 - \left[\frac{10[10+1]}{2} \right]^2 = \sum_{n=11}^{20} n^3$$

$$\left[\frac{20(21)}{2} \right]^2 - \left[\frac{10(11)}{2} \right]^2 = "$$

$$[210]^2 - [55]^2 = "$$

$$44100 - 3025 = "$$

$$\sum_{n=11}^{20} n^3 = 41075$$

$$\begin{array}{r} 44100 \\ - 3025 \\ \hline 41075 \end{array}$$

$\textcircled{1} \# \textcircled{21}$ How many numbers are divisible by 7 between 256 & 789.

$$259 \quad \dots \quad 784$$

$$n = \frac{\text{last term} - \text{first term}}{\text{common difference}} + 1$$

$$n = \frac{784 - 259}{7} + 1$$

$$\boxed{n = 76}$$

As we know that $S_1 = a_1$

Then, $S_2 = a_1 + a_2$ which can also be written as

$$S_2 = S_1 + a_2$$

$$S_2 - S_1 = a_2$$

So, $\boxed{a_n = S_n - S_{n-1}}$

Q #22 In an A.P. if 4th term is 14 & 9th term is 44. Find S_{23} ?

$$a_4 = 14$$

$$a_9 = 44$$

$$d = \frac{44 - 14}{5} = 30/5 = 6$$

$$a_{12} = 44 + 3(6)$$

$$\boxed{a_{12} = 62}$$

Now, $62(23) = 1426$

$$\text{middle term} = \frac{23+1}{2} = 12$$

$$\begin{array}{r} 3420 \\ -16 \\ \hline 3404 \end{array}$$

Q #23 if $a_1 = 40$, $a_3 = 16$, S_{22} ?

As 22 is even so we have two middle terms.

$$\text{M.T. Term}_1 = \frac{n}{2} = \frac{22}{2} = 11$$

& the other middle term will ahead of it
= 12

$$S_n = n \times \left[\frac{\text{MT}_1 + \text{MT}_2}{2} \right]$$

$$a_{11} = 40 + 2(4) = 40 + 8 = 48$$

$$a_{12} = 40 + 3(4) = 40 + 12 = 52$$

$$S_{22} = 22 \times \left[\frac{48 + 52}{2} \right] = 11(100) = 1100$$

$$d = \frac{40 - 16}{2} = \frac{24}{2} = 12$$

$$\begin{array}{r} 48 \\ + 52 \\ \hline 100 \end{array}$$

Q #24 IF first term of a G.P is 7 & L.T = 448 & Sum of n^{th} term 889

Find common ratio?

$$a_1 = 7$$

$$a_n = 448$$

$$S_n = 889$$

$$r = ?$$

$$S_n = \frac{a_1(r^n - 1)}{r - 1}$$

$$r - 1$$

$$889 = \frac{7(r^n - 1)}{r - 1}$$

$$889(r - 1) = 7(r^n - 1)$$

$$889r - 889 = 7r^n - 7$$

$$441r = 882$$

$$r = \frac{882}{441}$$

$$\boxed{r = 2}$$

In Geometric Progression:

For Finding Sum = We have two Shortcut formulas when we have first & last term.

$$\text{When } r > 1 \quad S_n = \frac{r(\text{last term}) - \text{First term}}{r - 1}$$

$$\text{When } r < 1 \quad S_n = \frac{\text{First term} - r(\text{last term})}{1 - r}$$

①*② If p^{th} term of an H.P is q & q^{th} term is p . Find $(p+q)^{\text{th}}$ term.

$$\left. \begin{array}{l} a_p \rightarrow q \\ a_q = p \end{array} \right\} \text{H.P.}$$

$$\left. \begin{array}{l} a_p = \frac{1}{q} \\ a_q = \frac{1}{p} \end{array} \right\} \text{A.P.}$$

$$d = \frac{\frac{1}{p} - \frac{1}{q}}{q - p} = \frac{\frac{q-p}{pq}}{\frac{q-p}{1}} = \frac{1}{pq}$$

$$a_{p+q} = a_p + (p+q-p) \left(\frac{1}{pq} \right)$$

$$a_{p+q} = \frac{1}{q} + (q) \left(\frac{1}{pq} \right)$$

$$a_{p+q} = \frac{1}{q} + \frac{1}{p} = \frac{p+q}{qp}$$

$$H_{p+q} = \frac{qp}{p+q} \quad \left. \right\} \text{in H.P.}$$

When we have to find Sum of Series & Means are given in Arithmetic Progression we will use the formula $\frac{(n+2)(a+b)}{2}$.

①*② Find Sum of Series when 6 arithmetic means are inserted between 10 & 40.

$$\frac{(6+2)(10+40)}{2}$$

$$8/2 (50)$$

$$50(4)$$

$$\boxed{200}$$

Product of n Geometric Mean:

$$\text{Product of } n \text{ geometric mean} = (ab)^{n/2}$$

Q#(2) Find Product of 3 geometric mean between 3 & 243.

$$\begin{aligned} & \therefore (3 \times 243)^{3/2} \\ & = (729)^{3/2} \\ & = (3^6)^{3/2} \\ & = 3^9 \end{aligned}$$

Note: The mean between Arithmetic & Harmonic is called Geometric mean

Q#(3) If Arithmetic mean, Geometric mean & Harmonic mean are satisfying the relation $2A + G^2 = 27$ & if the H-M between 2 numbers is 4 then the numbers are?

$$2A + G^2 = 27$$

$$\therefore G^2 = AH$$

$$2A + AH = 27$$

$$2A + 4A = 27$$

$$A = 27/6 = 9/2$$

$$\boxed{A = 9/2}$$

$$2(9/2) + G^2 = 27$$

$$G^2 = 27 - 9$$

$$G^2 = 18$$

$$\boxed{G = \sqrt{18}}$$

You can also check by putting values in equation $2(9/2) + 18 = 27$
 $27 = 27$

Note:

$$\text{Q#(4) } \textcircled{a} A.M > G.M > H.M$$

Assume two extreme no's of your choice

$$\textcircled{b} A.M > H.M > G.M$$

Let $a=2, b=10$

$$\textcircled{c} G.M > H.M > A.M$$

$$A.M = 6$$

$$\textcircled{d} H.M > A.M > G.M$$

$$G.M = \sqrt{20} = 4.4$$

$$H.M = 3.3$$

For Finding Square root Trick:

$$\hookrightarrow A.M > G.M > H.M$$

$$\leftarrow \sqrt{20} \rightarrow 4.4$$

but $A.M < G.M < H.M$ when $G < 0$

As we have perfect square check

i.e. 16

Now, Subtract root value from that perfect square value

$$20 - 16 = 4$$